Arista Access Node HEB-ACC-25 Hardware/Environmental Fault Resolution

Node: HEB-ACC-25 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1957

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve hardware/environmental faults, specifically ASIC memory parity errors, line card crashes, and power supply redundancy loss on the Arista Access Node HEB-ACC-25 at the Hebbal Edge site. The Access Node is a critical component of the network infrastructure, responsible for providing Layer 2 and Layer 3 access services to end-users. The Arista 7150 series Switch, used in this node, features a modular architecture with a combination of hardware and software components. The ASIC, or Application-Specific Integrated Circuit, is a critical component responsible for performing switching and routing functions. However, due to its complex design and high-speed operations, the ASIC is prone to errors, such as memory parity errors, which can cause the line card to crash. Additionally, the power supply redundancy loss can occur due to a hardware failure or environmental factors, such as power fluctuations or overheating. The combination of these factors can lead to network instability and downtime. This SOP aims to provide a structured approach to diagnosing and resolving these issues, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Hebbal Edge site for physical access to the switch.

- Verify the redundant link status to ensure the network remains operational during the diagnostic and resolution process.

- Check the power supply unit (PSU) status and ensure it is functioning properly.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

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Command: show platform, Expected Output: Platform information, including version and revision information, Description: This command provides a detailed view of the switch's hardware and software configuration, including the ASIC version and revision information, which can help identify potential issues.
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Command: show system internal error, Expected Output: Error information, including memory parity errors and line card crashes, Description: This command provides a detailed view of the system's internal error information, including memory parity errors and line card crashes, which can help identify the root cause of the issue.

Command: show power status, Expected Output: Power supply unit (PSU) status information, Description: This command provides a detailed view of the PSU status, including whether it is functioning properly or experiencing issues.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: no switch all, Description: This command powers off the switch and allows for the replacement of the faulty ASIC or line card. Ensure that the replacement part is properly seated and secured before proceeding.

Step 2: Step: Verify the power supply unit (PSU) status, Command: show power status, Description: This command verifies the PSU status and ensures it is functioning properly. If the PSU is faulty, it should be replaced before proceeding.

Step 3: Step: Reboot the switch, Command: switch all, Description: This command powers on the switch and brings it back online. Ensure that all components are functioning properly before proceeding.

Post-Check Verification

```
Command: show system internal error, Expected Output: No error information, Description: This command verifies that there are no internal errors or issues with the system.

Command: show power status, Expected Output: Power supply unit (PSU) status information, indicating proper functioning, Description: This command verifies that the PSU is functioning properly and there are no issues with power supply.

Command: show platform, Expected Output: Platform information, including version and revision information, confirming proper functioning, Description: This command verifies that the switch's hardware and software configuration is proper and functioning as expected.
```

Roll-back Procedures

- Step: Revert the switch to its previous configuration, Command: copy running-config startup-config, Description: This command reverts the switch to its previous configuration, reversing any changes made during the diagnostic and resolution process.

- Step: Verify the system's internal error information, Command: show system internal error, Description: This command verifies that there are no internal errors or issues with the system.

- Step: Verify the power supply unit (PSU) status, Command: show power status, Description: This command verifies that the PSU is functioning properly and there are no issues with power supply.